Element F: Consideration of Design Viability

What might be difficult or require extra time/planning for your solution?

- Ensuring material stability and long-term durability with recycled materials may be difficult.
- Sourcing uniform materials like plastic and wood could take extra planning and coordination
- Perfecting the triangular support angle to prevent tipping might require multiple prototypes.
- Keeping the final design under a very strict cost limit while the design still needs to remain both functional and attractive could also be challenging.

What potential obstacles (materials, process, timing, etc.) might you face?

- Inconsistent design, size, or quality of recycled materials
- Difficulty cutting or shaping wood with available tools and materials.
- Materials not adhering properly during construction.
- Design slipping or tipping over during actual book use.
- Staying within budget
- Limited time to test multiple prototypes.

How do you plan to address them?

- For inconsistent material sizes, we will pre-sort and measure all materials before cutting.
- For difficulty with cutting tools, we will use pre-cut wood scraps.
- To ensure materials adhere properly, we'll test and choose the strongest adhesive and allow proper drying time.
- To prevent slipping and tipping, we will attach rubber dots or grips and test the design under real shelf conditions.
- For cost issues, we will use fully recycled materials and document estimates.
- We will build early test versions to avoid last minute fixes and allow for stakeholder feedback.

Written Context + Supporting Justifications

- We may have to test different adhesives and reinforcement methods because wood and plastic don't always bond well, which means our bookend might fall apart under pressure. We need to prioritize strucutrual integrity in recycled designs.
- We will address the sippage issue by doing a rubber base test with recycled stoppers. Non-slip grip is one of our top priorities.

- Lastly, In order to complete/ensure the design remains affordable and functional, we will need to use only recycled or donated materials, which will address the school's sustainability goals and cost constraints.

How will you know that your end users will be satisfied?

- When we consulted Ms. Tatro, our primary stakeholder, who emphasized the importance of durability, low cost, and non-slippage. This confirms that our current direction was aligned with the needs of the ERHS library.

Obstacle + Design Viability Table.

Design Challenge or Obstacle	**Connection to Design Requirement**	**Plan to Adress It/Credible Evidence**
Inconsistent Material Sizes.	Must be uniform and reliable to build durable bookend	Pre-measure and sort all recycled materials.
Difficult wood shaping.	Must be easy and realistic to construct	Use Pre-cut wood scraps from shop/donations.
Weak adhesion between materials	Must stay intact under pressure/weight	Test Adhesives, clamp materials overnight.
Design slipping	Must remain stable on shelf with heavy books.	Add rubber grips to base, test for friction.
Staying under cost limit.	Must cost $0. Only recycled materials used	Reuse materials from home, school, and donations.
Prototype Timing	Must be tested and built within project timeline	Build early, test quickly.
Supporting Atleast 339.8g	Must hold weight without collapsing	Structural testing with books exceeding the weight.
Meeting Stakeholder needs.	Must align with library goals for stability, safety, and sustainability.	Regular feedback and review with Ms.Tatro.
Sustainable Materials.	Must be environmentally safe.	100% recycled, no new purchases